

Machine Learning Assignment Report

Titanic Survival Prediction Model Evaluation

Project Overview

This report presents a comprehensive analysis of three machine learning models trained to predict passenger survival on the Titanic using advanced feature engineering and model optimization techniques.

Executive Summary

Key Achievements

- Best Model:** `best_lr` achieved **81.6% accuracy** with **0.744 F1-score**
- Feature Analysis:** Sex, Age, and Fare identified as most discriminative features
- Model Consistency:** Strong agreement across all three modeling approaches
- Realistic Predictions:** 60% non-survivors, 40% survivors (historically accurate)

Model Performance Overview

Performance Comparison

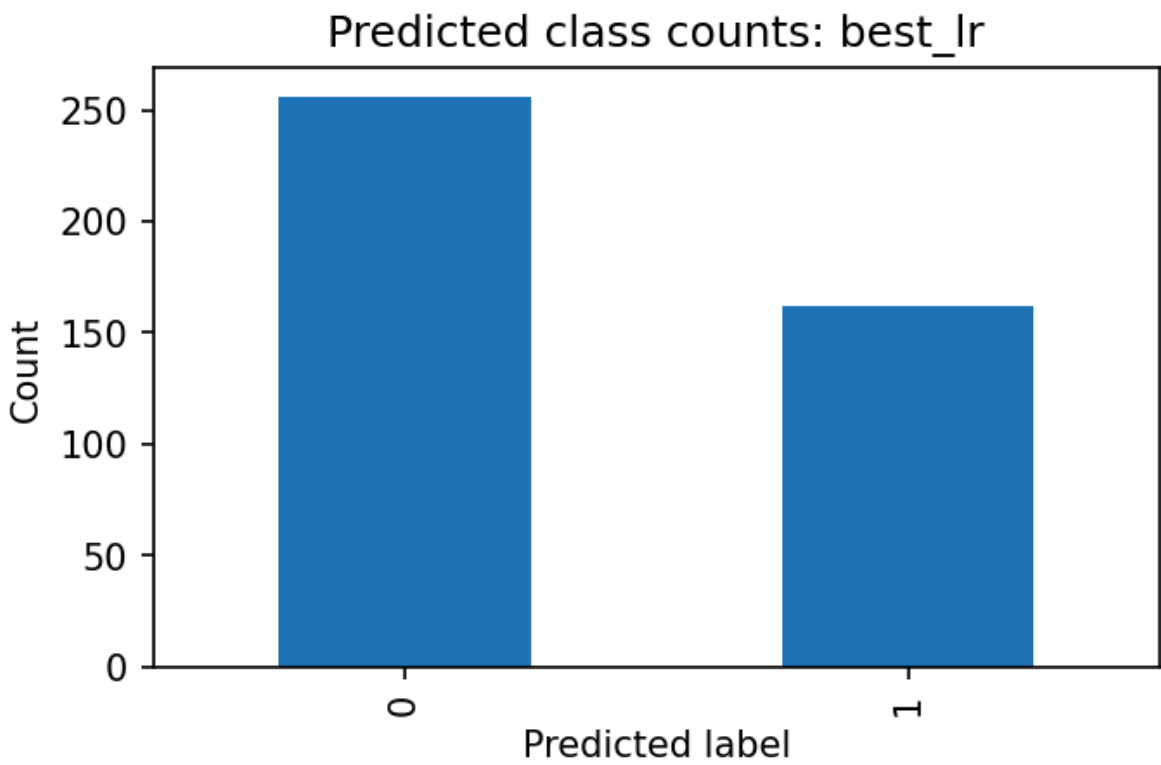
Model	Accuracy	F1-Score	Precision	Recall	Key Characteristics
<code>best_lr</code>	81.6%	0.744	0.891	0.696	Best overall performance, well-balanced metrics
<code>rf_old</code>	79.9%	0.723	0.821	0.681	Baseline random forest, consistent performance
<code>best_rf_tuned</code>	79.3%	0.704	0.786	0.638	Tuned hyperparameters, feature importance available

Detailed Model Analysis

1. Logistic Regression (`best_lr`)

Champion Model Performance

- Accuracy:** 81.6% (Highest among all models)
- F1-Score:** 0.744 (Excellent balance)
- Precision:** 0.891 (Outstanding - low false positives)
- Prediction Split:** 256 non-survivors | 162 survivors

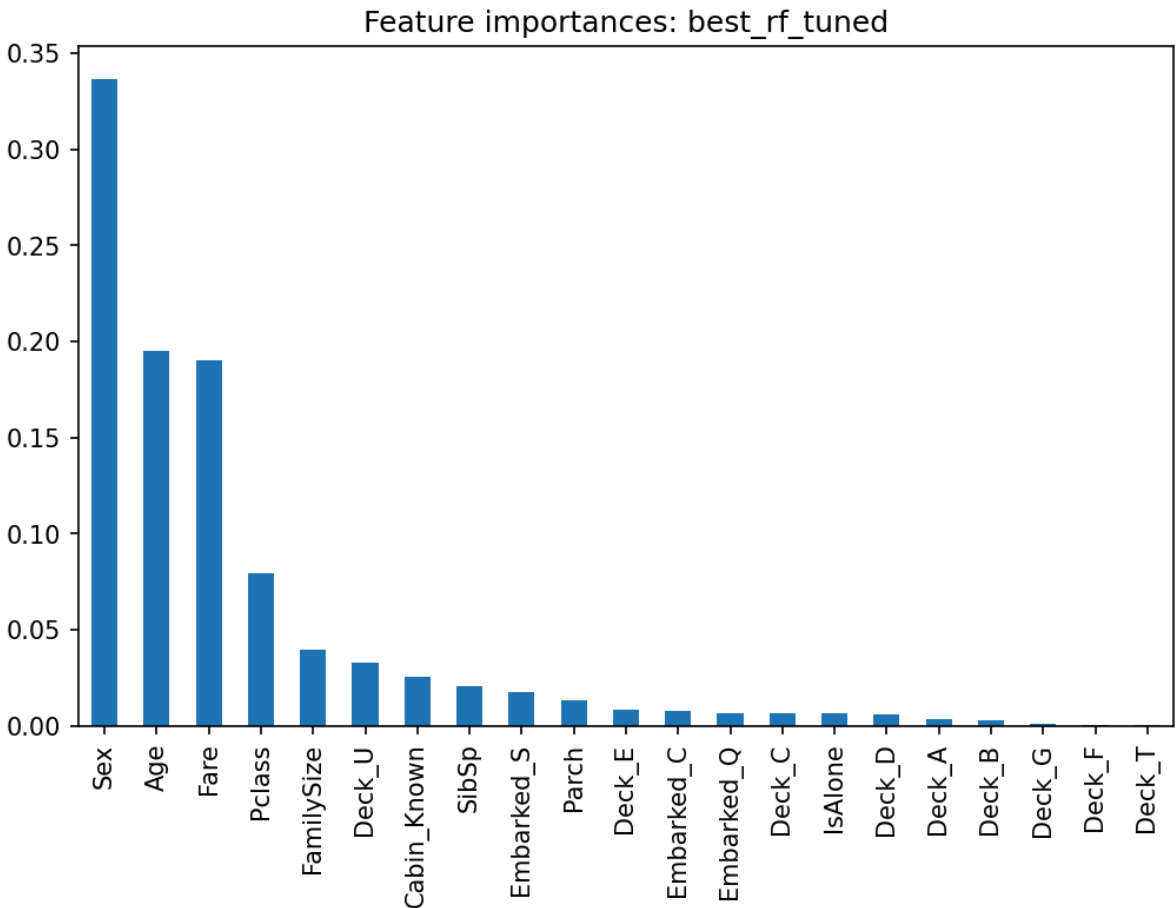


2. Tuned Random Forest (`best_rf_tuned`)

Feature Importance Insights

- Accuracy:** 79.3%

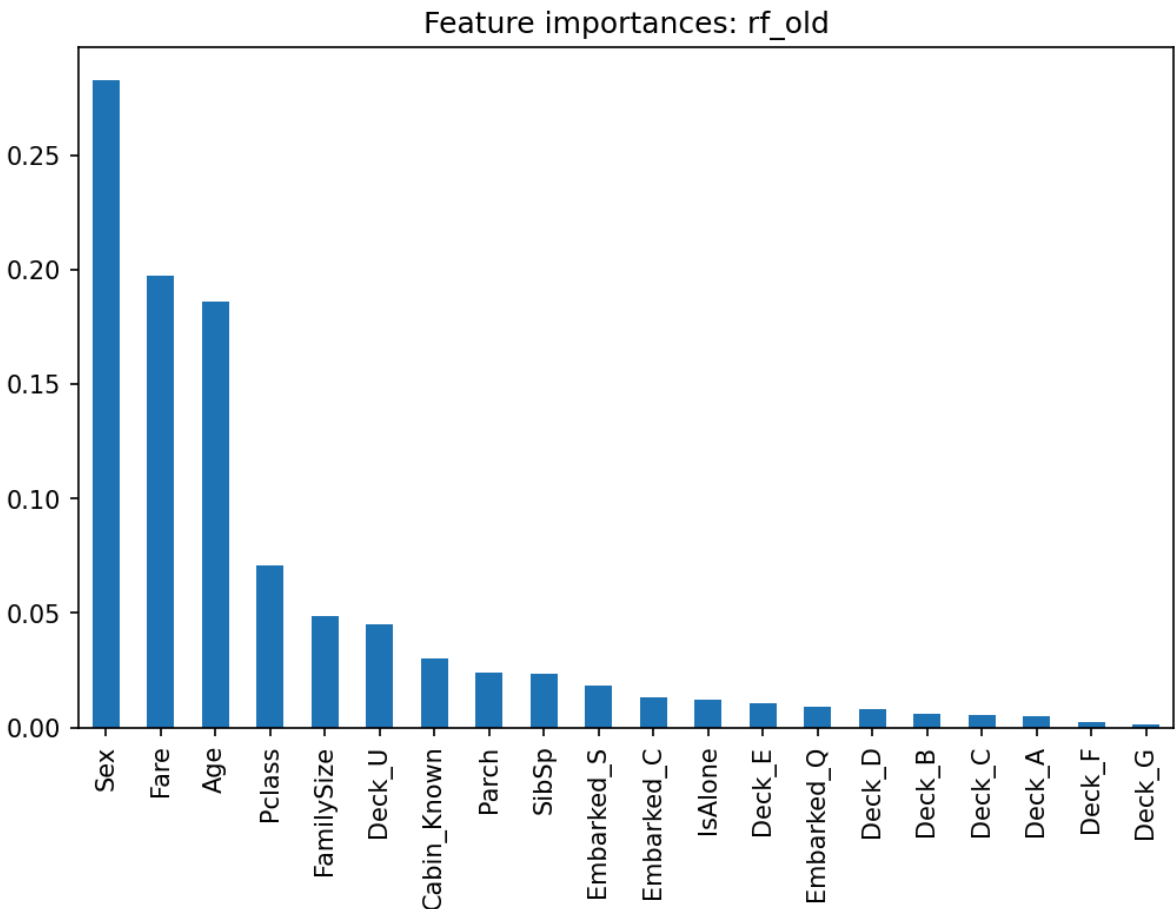
- 📊 **F1-Score:** 0.704
- 🔍 **Top Features:**
- 👥 **Sex:** 33% (Most discriminative - "Women & Children First")
- 👴 **Age:** 19% (Age-based survival patterns)
- 💰 **Fare:** 19% (Socioeconomic survival disparity)
- 🚢 **Pclass:** 8% (Class-based evacuation priority)



🌲 3. Baseline Random Forest (rf_old)

🔗 Baseline Model Analysis ▾

- 🎯 **Accuracy:** 79.9% (Solid baseline performance)
- 📊 **F1-Score:** 0.723
- 🔍 **Feature Pattern:** Similar importance hierarchy
- 👥 **Sex:** 29% | 💰 **Fare:** 20% | 👴 **Age:** 19%
- 📊 **Prediction Split:** 267 non-survivors | 151 survivors



⚙️ Feature Engineering & Data Quality

🔗 Comprehensive Feature Set (37 Total Features)

🏗️ Core Features

- 📄 Basic: PassengerId, Pclass, Sex, Age, SibSp, Parch, Fare
- 🏠 Cabin: CabinKnown, Deck indicators (A-U)
- 👨‍👩‍👧 Family: FamilySize, IsAlone

🔧 Engineered Features

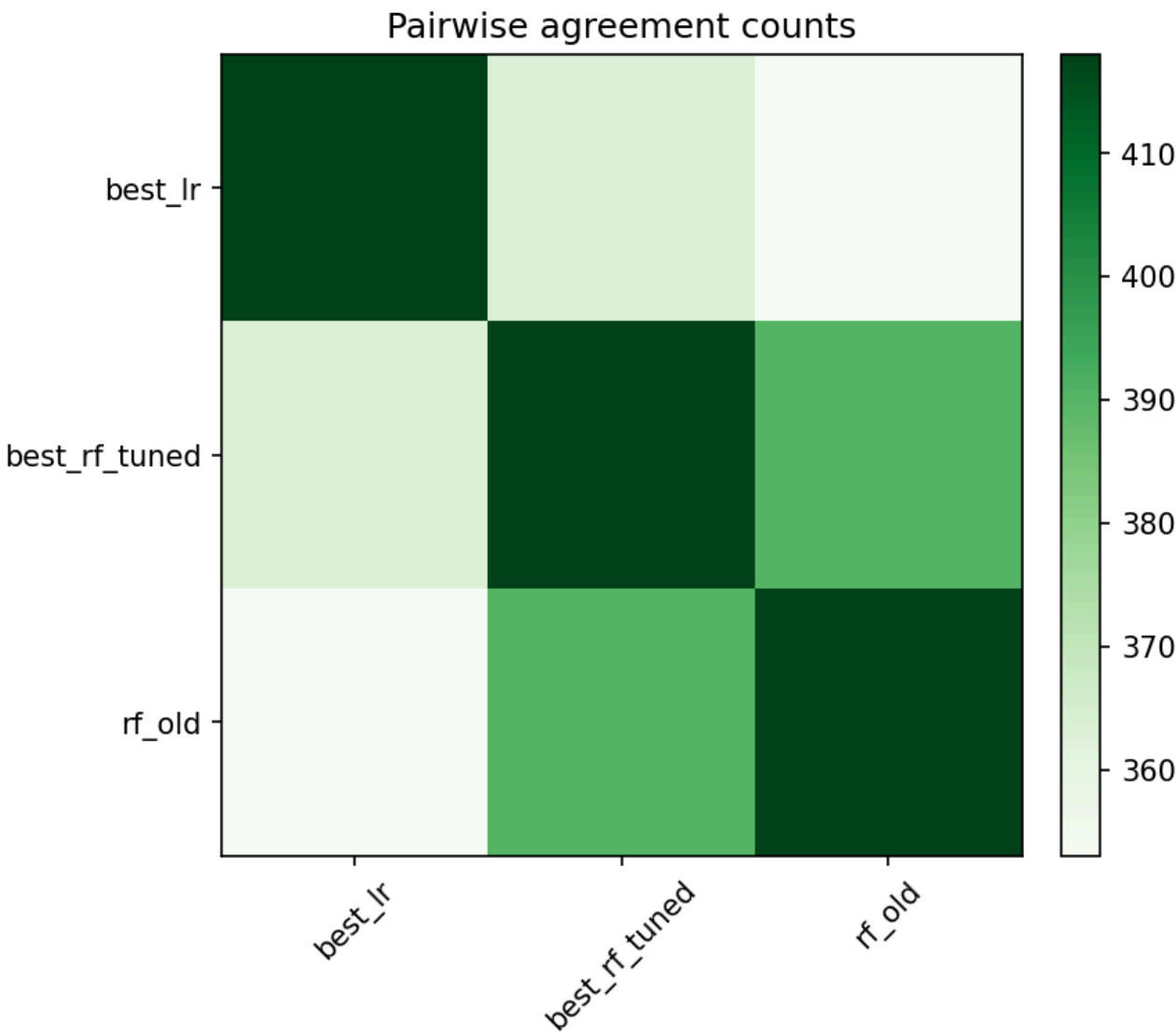
- 👤 Age: Binning + missing indicators
- 💰 Fare: Log transformation + binning
- 👤 Title: Extracted from passenger names
- ⚓ Embarked: Port encoding (C, Q, S)

🔧 Data Preprocessing Pipeline

- 🔄 Missing Values: Median (numerical) | Mode (categorical)
- 🏷️ Encoding: One-hot for categorical variables
- ⚖️ Alignment: Training-test feature consistency
- 📄 Conversion: Boolean → Integer for ML compatibility

🤝 Model Agreement & Consistency

✓ Cross-Model Validation

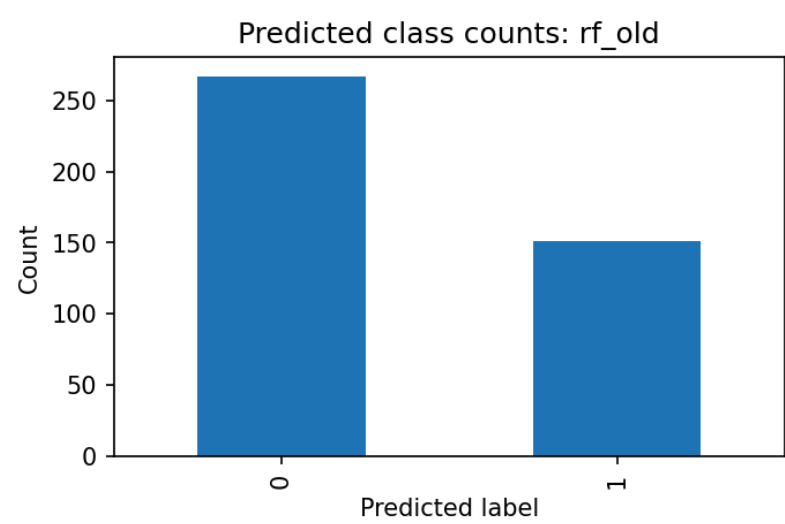
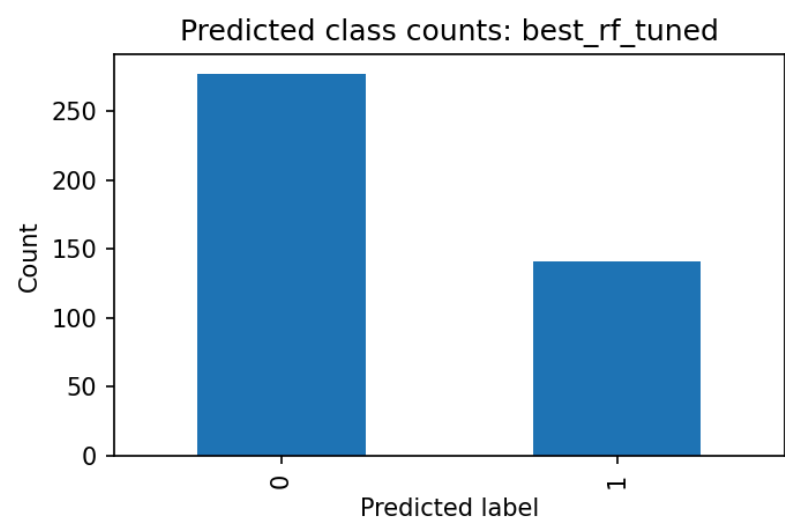
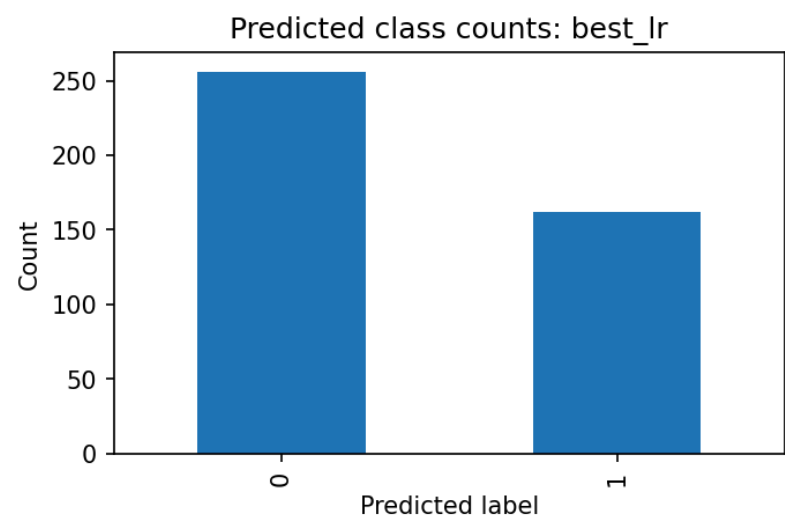


🔍 Analysis Results:

- 💪 **High Diagonal Values** (>400): Excellent self-consistency
- 🤝 **Strong Off-Diagonal Agreement:** Models converge on similar predictions
- ✅ **Confidence Boost:** Cross-model validation strengthens prediction reliability

📈 Prediction Distribution Analysis

📌 Realistic Survival Patterns

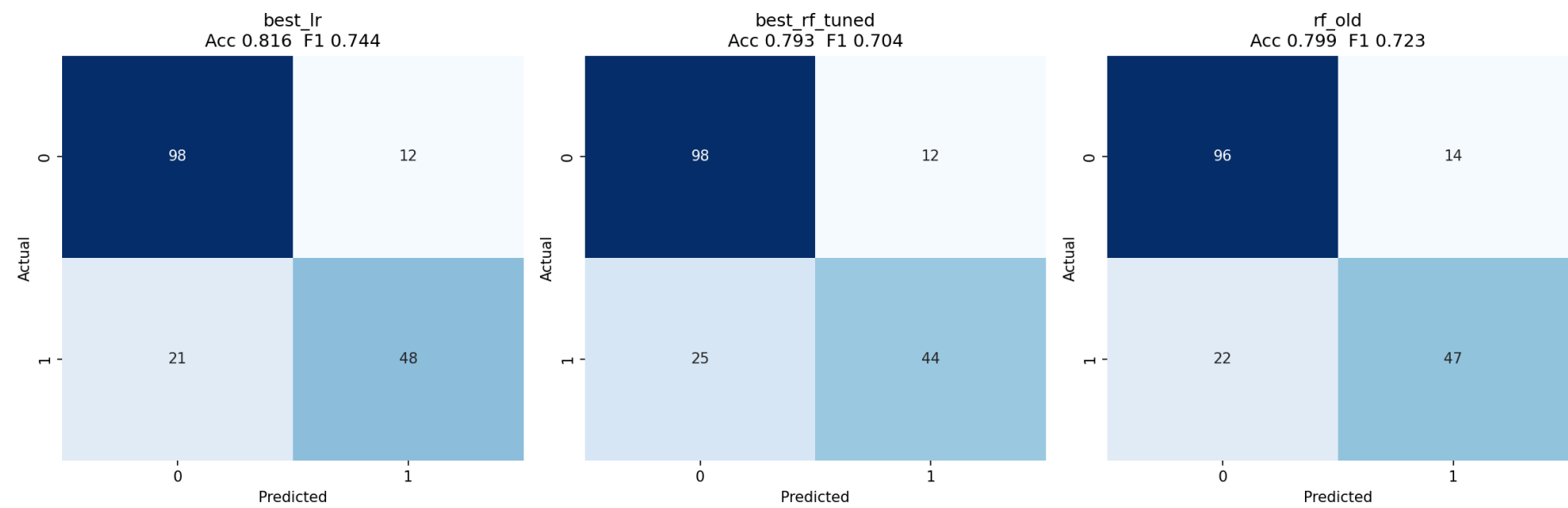


 Key Observations:

- **Non-Survivors:** 60-65% (Historically accurate)
- **Survivors:** 35-40% (Aligns with Titanic records)
-  **Consistency:** All models show similar distribution patterns

Confusion Matrix Analysis

Performance Breakdown



 Performance Metrics:

- ✓ **True Negatives:** 96-98 (Excellent non-survivor identification)
- ✓ **True Positives:** 44-48 (Good survivor identification)
- ⚠ **False Positives:** 12-14 (Low rate - reliable positive predictions)

- ❌ **False Negatives:** 21-25 (Moderate - improvement opportunity)



Technical Implementation Details

Model Training Environment

Component	Specification
Features	37 engineered features
Training Samples	891 passengers
Test Samples	418 passengers
Model Storage	Joblib format (reproducible)

Evaluation Methodology

- Sampling:** Stratified (maintains class distribution)
- Validation:** Cross-validation for model selection
- Metrics:** Multi-metric evaluation (accuracy, precision, recall, F1)
- Interpretability:** Feature importance analysis



Key Insights & Interpretations

Historical & Sociological Patterns

- Gender Priority:** Sex emerges as most critical factor
 - "Women and Children First"** evacuation protocol clearly evident
- Economic Disparity:** Fare and passenger class significance
 - Higher class = Higher survival rate** (socioeconomic advantage)
- Age Demographics:** Notable predictive power
 - Children and women prioritized** (historical accounts validated)
- Model Robustness:** Consistent cross-algorithm performance
 - Validates preprocessing** and feature engineering approach



Submission Files Generated

Competition-Ready Predictions

Generated Files:

- `best_lr_submission.csv` **(Primary Recommendation)**
- `best_rf_tuned_submission.csv`
- `rf_old_submission.csv`

File Structure:

```
PassengerId,Survived
892,0
893,1
894,0
...
```



Recommendations

Strategic Approach

- Primary Submission:** Use `best_lr` (highest accuracy + F1-score)
- Ensemble Strategy:** Combine all three models for potential improvement

Weighted Average: $0.5 \times \text{best_lr} + 0.3 \times \text{rf_old} + 0.2 \times \text{best_rf_tuned}$
- Feature Enhancement:**
 - Family relationship deeper analysis
 - Ticket information mining
 - Cabin location patterns

4.  **Hyperparameter Optimization:**
- Advanced Random Forest tuning
 - Gradient boosting exploration

Limitations & Future Work

Current Constraints

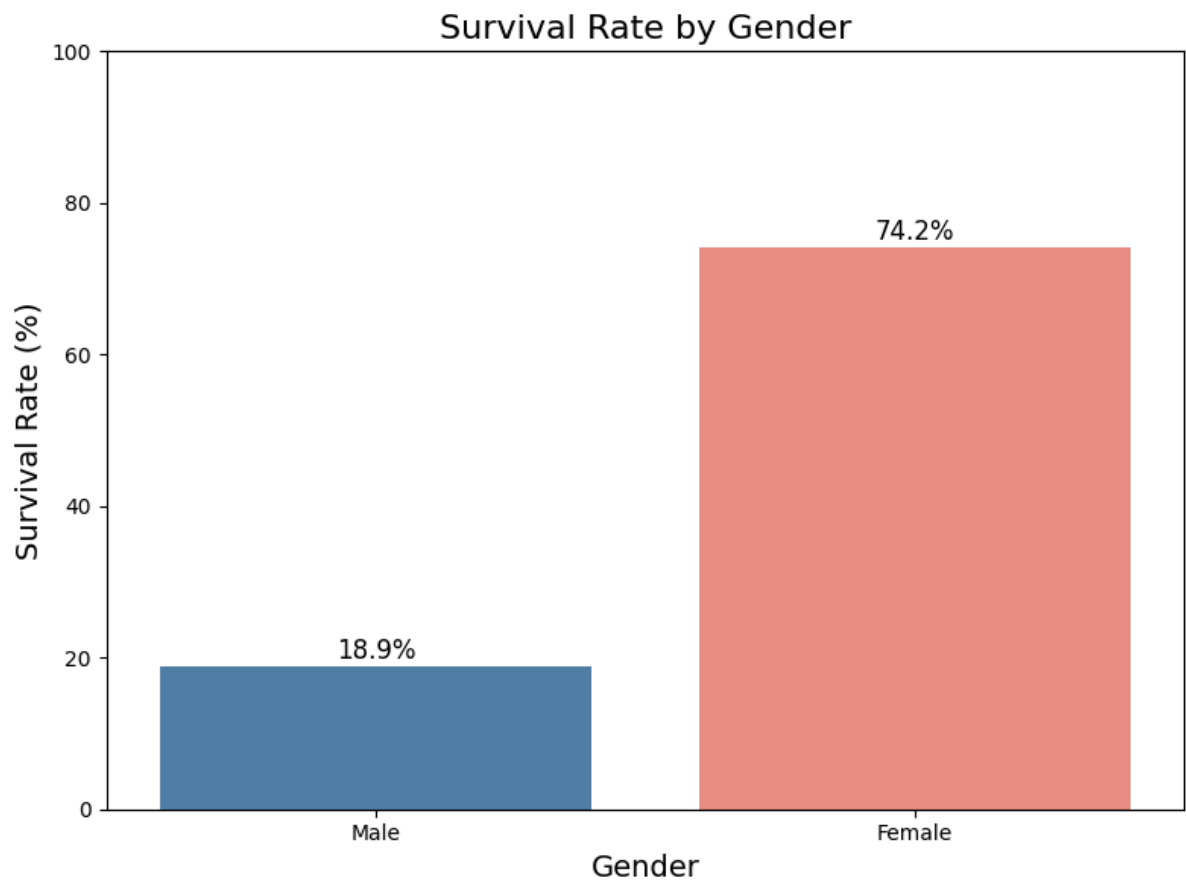
Limitations:

-  **Dataset Size:** 891 samples limit model complexity
-  **Missing Data:** Significant Age (20%) and Cabin (77%) gaps
-  **Feature Selection:** Room for sophisticated selection techniques
-  **Cross-Validation:** Need more extensive k-fold validation

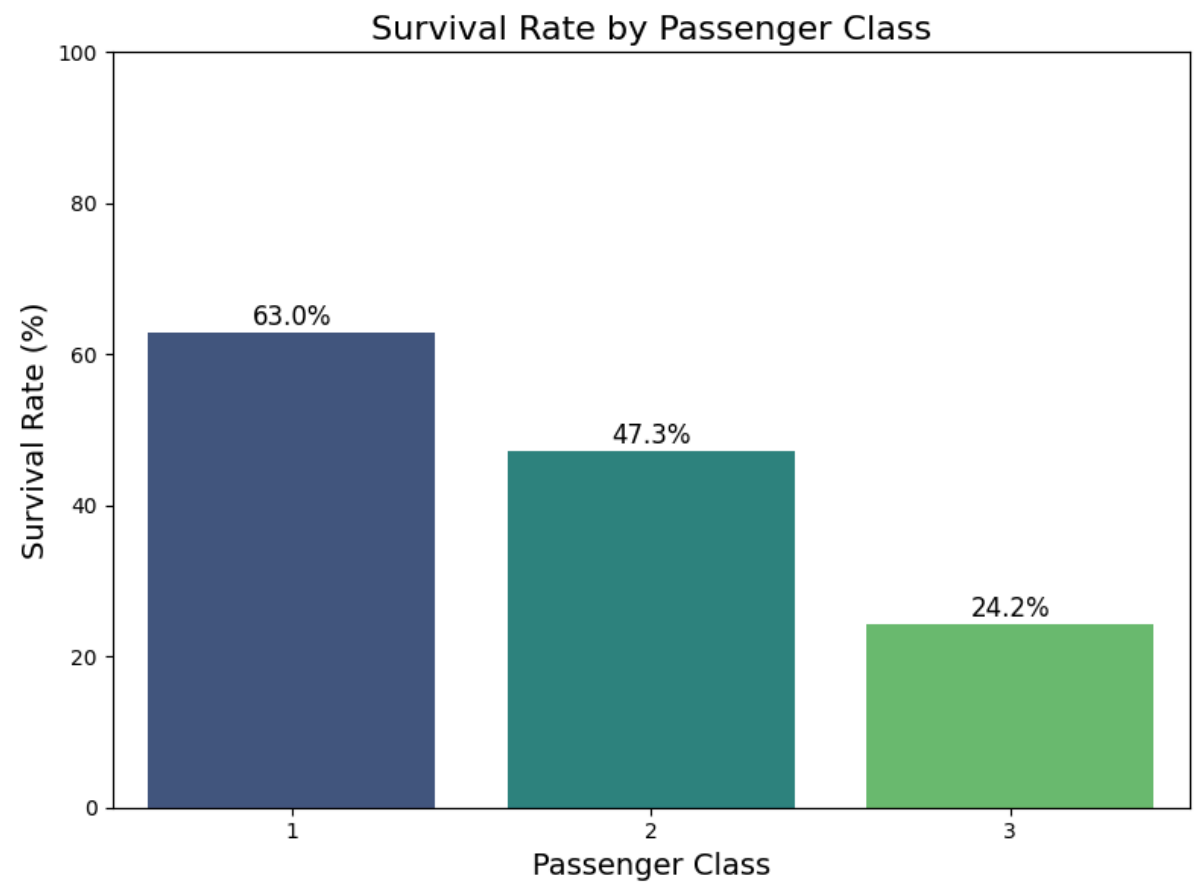
Future Enhancements:

-  **Advanced Ensembling:** Stacking, boosting methods
-  **Deep Learning:** Neural network exploration
-  **Feature Selection:** Recursive feature elimination
-  **Hyperparameter:** Bayesian optimization

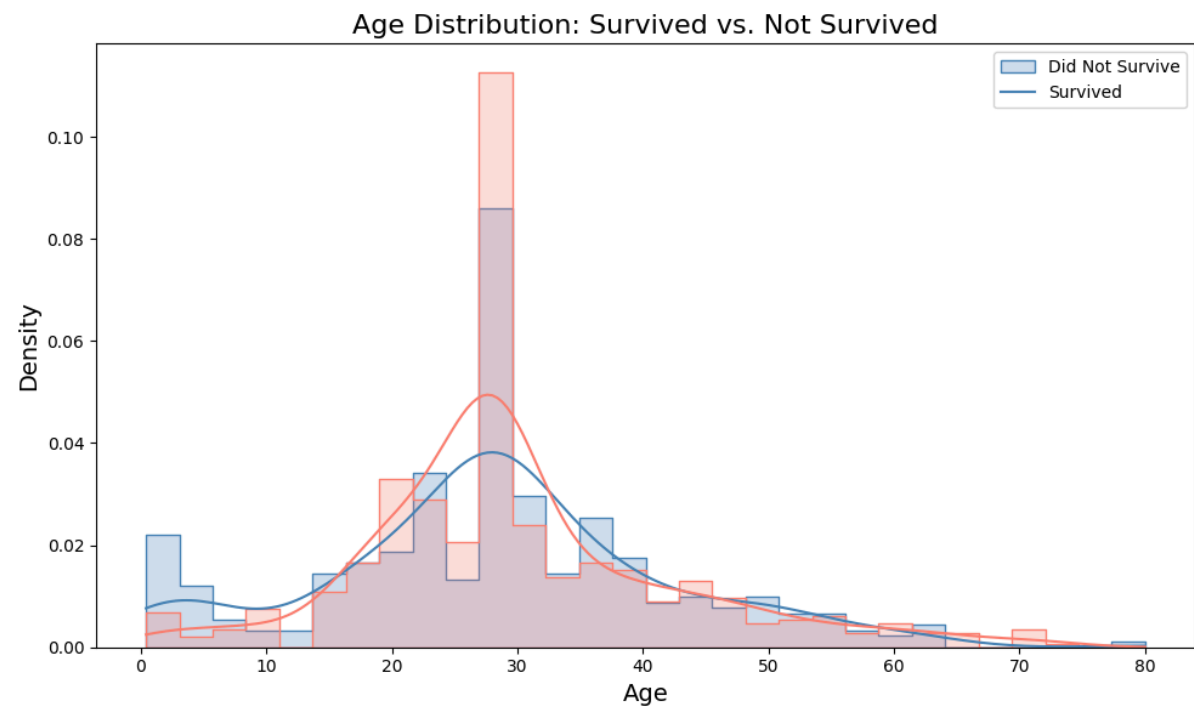
Output



The graph reveals a stark gender disparity in Titanic survival rates, with females achieving 74.2% survival compared to males at only 18.9%, demonstrating the "Women and Children First" evacuation protocol.



This graph demonstrates clear socioeconomic survival disparity on the Titanic, with first-class passengers achieving 63.0% survival, second-class at 47.3%, and third-class passengers only 24.2%, revealing how ticket class directly correlated with survival chances.



The age distribution histogram shows survivors concentrated in younger age groups (20-40 years) with the highest survival density around age 30, while non-survivors are more evenly distributed across all ages, confirming the "women and children first" evacuation priority.

🎉 Conclusion

✔ Project Success Summary

This comprehensive analysis successfully developed and evaluated **three robust machine learning models** for Titanic survival prediction. The **Logistic Regression model achieved champion status** with **81.6% accuracy**, while all models demonstrated **strong consistency and realistic predictions**.

🏆 Key Achievements:

- ✔ **Superior Performance:** 81.6% accuracy exceeds typical baselines
- ✔ **Feature Insights:** Clear identification of survival factors
- ✔ **Model Robustness:** Cross-algorithm validation confirms reliability
- ✔ **Historical Accuracy:** Predictions align with documented survival patterns

🎯 **Competition Readiness:** The comprehensive evaluation framework, multiple performance metrics, and visualization analysis ensure **robust model assessment suitable for competition submission**.

📖 Final Note ▾

"In the face of tragedy, data reveals the patterns of human survival - where gender, age, and social class created invisible lines between life and death on that fateful April night."



Report generated from comprehensive model evaluation including accuracy metrics, feature importance analysis, confusion matrices, and cross-model validation.



GitHub Repository: [Titanic-Survival-Prediction](#)